

EviCoord: reusable coordination core + domain adapter

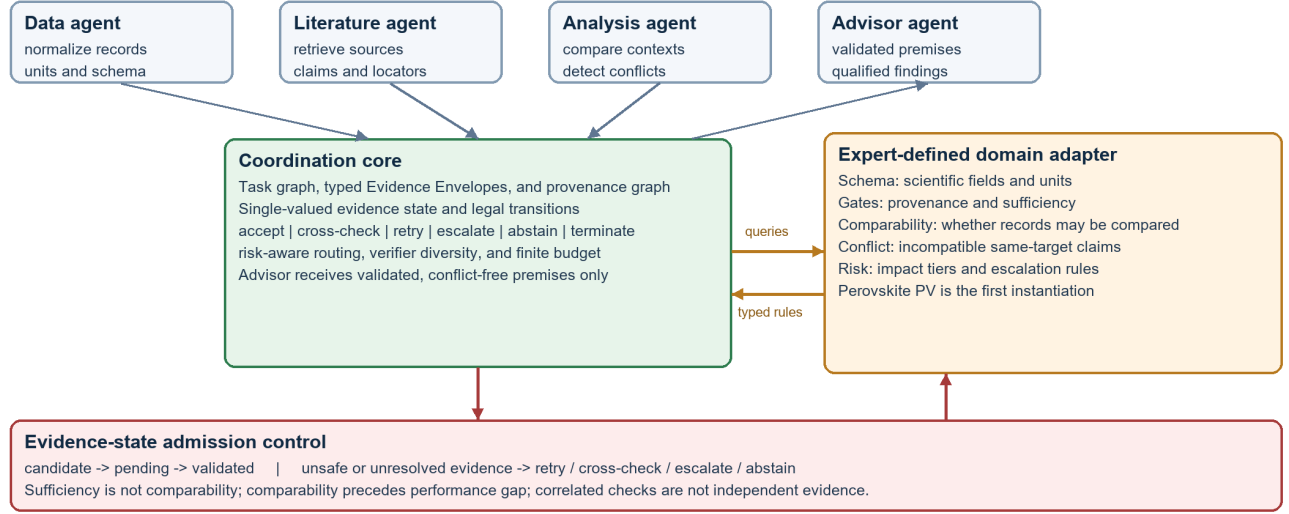


Figure 1: EviCoord separates a reusable evidence-state coordination core from an expert-defined domain adapter. Agents are capability boundaries and need not all be language models. The adapter determines what counts as traceable, sufficient, comparable, conflicted, or high risk; the core governs legal evidence transitions and resource-bounded actions.

that is, the outcome gap is a partial function evaluated only after admissibility passes; its value never rescues an inadmissible pair.

3.2 Core-Adapter separation

The reusable coordination core is

$$C = \langle G_T, E, G_P, U, B, H, T \rangle, \quad (5)$$

where E is the envelope store, G_P the provenance graph, U the action set, B the remaining budget, H the action history, and T the transition function. Actions are

$$U = \{\text{DELEGATE, ACCEPT, CROSS-CHECK, RETRY, ESCALATE, ABSTAIN, TERMINATE}\}.$$

A coordination policy is a map $\pi : S \rightarrow \Delta(U)$ over states induced by the core and adapter. Let I_π^{unsafe} mark an unsafe emitted output. For a required clean-evidence coverage κ , one risk-cost operating point can be written

$$\begin{aligned} \min_{\pi} \quad & \mathbb{E} \left[w_r I_\pi^{\text{unsafe}} + \lambda C_\pi \right] \\ \text{s.t.} \quad & \Pr(\text{emit} \mid \text{clean}) \geq \kappa, \quad N_\pi \leq B. \end{aligned} \quad (6)$$

EviCoord is a rule-specified feasible policy, not a claimed optimizer for an unknown field distribution; we therefore report risk, coverage, and cost separately. The domain adapter is

$$\mathcal{A}_d = \langle \Sigma_d, \text{Gates}_d, \text{Comp}_d, \text{Conflict}_d, \text{Risk}_d \rangle. \quad (7)$$

Σ_d types scientific fields, units, and claim-specific provenance requirements. Gates_d defines schema, traceability, and sufficiency predicates. Comp_d and Conflict_d define pairwise admissibility and

same-target incompatibility. We require SevereConflict_d to be pairwise symmetric; a nonsymmetric adapter must evaluate both directions. Risk_d maps tasks and observed evidence states to impact tiers and escalation rules. Changing adapters does not change the transition program, but it does require new expert validation; interface reuse does not imply transferred empirical accuracy.

Define

$$\begin{aligned} \text{Bad}_d(x, e; E) = & \neg \text{Schema}_d(e) \vee \neg \text{Traceable}_d(e) \\ & \vee \neg \text{Sufficient}_d(e) \vee \neg \text{Admissible}_d(x, e) \\ & \vee \text{SevereConflict}_d(e, E \setminus \{e\}). \end{aligned} \quad (8)$$

Traceability is claim-type specific. For a claim about a versioned database record, a snapshot hash, record identifier, DOI, and transformation path may constitute the source. For a claim attributed to a paper, validation additionally requires a page, table, figure, or text-span locator. A DOI alone identifies a candidate document, not where the scientific statement occurs.

3.3 Hierarchical coordination policy

Algorithm 1 implements the protocol used in our evaluation. Hard predicates precede reliability scores. Schema or provenance failures trigger bounded repair; insufficient or incomparable evidence is contained; and every severe conflict triggers a structurally diverse cross-check. If that conflict remains unresolved, a high-impact task is escalated and a low-impact task is abstained from. A cross-check is structurally diverse only if it changes at least one of the producer, parsing tool, or evidence source. Repeated calls to the same source family count as correlated votes, not independent evidence. Only evidence passing all required gates and whose